

AI for Marketing Decisions

Course: AI for Marketing Decisions

Specialty: AI in Business

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<https://github.com/ilyas06089-collab/Ilyas/upload>

1. Executive Summary

In the contemporary hyper-competitive corporate ecosystem, data has transformed from a structural asset into the primary engine of strategic growth. Modern marketing operations can no longer survive on executive gut instinct, historical inertia, or broad, static demographic assumptions. To optimize customer acquisition, maximize long-term retention, and protect corporate profit margins, global enterprises like Amazon, Spotify, and Coca-Cola rely heavily on automated algorithms and data-driven intelligence systems.

This comprehensive final exam project establishes an end-to-end AI-driven marketing decision system built entirely from empirical client data. By deeply connecting specialized automated processes in Python with dynamic business intelligence layouts in Power BI, this framework addresses five core pillars of high-performance marketing: customer behavioral profiling, predictive risk mitigation, prescriptive personalization, sentiment intelligence, and ethical machine learning compliance.

The system engineered in this study bridges the operational gap between raw, messy database entries and executive decision-making. The core analytical engine performs statistical data cleaning, engineers crucial financial indicators, executes machine learning behavior classification, maps personalized product affinity scores, and processes unstructured social media feedback using Natural Language Processing (NLP).

The final delivery mechanism manifests as a fully integrated, interactive 8-page Power BI Marketing Dashboard designed specifically for a Chief Marketing Officer (CMO). This interface provides automated, single-source-of-truth answers regarding performance metrics, channel ROI, customer lifecycle vulnerabilities, and algorithmic

fairness metrics, allowing senior leadership to make high-impact campaign resource reallocations in under ten minutes.

2. Project Architecture & Relational Dataset Overview

2.1. System Architecture Pipeline

The technical design of the marketing intelligence system utilizes a modular, two-tier architecture:

1. **The Python Analytics Engine (Jupyter Notebook):** Acts as the data processing, statistical extraction, and machine learning model factory. It takes raw, unstructured information, executes transformation routines, runs analytical modeling algorithms, and exports optimized tabular structures.
2. **The Business Intelligence Presentation Layer (Power BI Desktop):** Serves as the relational semantic data model and interactive interface. It ingests the processed outputs, applies complex DAX financial metrics, establishes interactive query behaviors, and serves as the visual command center for the organization.

2.2. Relational Schema and Table Definitions

The data engine is built upon five specialized tables connected via a star-like semantic model schema, as validated by internal system design blueprints:

- **customer_analytics_clean (Central Fact Table):** Houses granular operational, biographical, and purchasing records for every client. Key variables include unique customer identifiers (**ID**), shipping performance metrics (**Reached_on_Time**), engagement scoring matrices (**Engagement_Score**), support contact touchpoints (**Customer_care_calls**), historical customer ratings (**Customer_rating**), and financial vectors (**Cost_of_the_Product**, calculated **AOV**, and calculated **CLV**).
- **segment_summary (Dimension Table):** An aggregated behavioral lookup table derived directly from machine learning segmentation models. It defines distinct consumer cohorts, tracking metrics like average cluster order value (**Avg_AOV**), projected lifetime values (**Avg_CLV**), customer counts (**Count**), and baseline campaign response ratios (**Camp_Resp**).
- **fairness_summary (Audit Layer Table):** A dedicated administrative evaluation matrix designed to ensure corporate algorithmic transparency and monitor bias across protected demographic subclasses. It compares actual historical metrics (**Actual_HVC**) against neural classification targets (**Predicted_HVC**) across specific sub-groups (**Group_Type**, **Group_Value**).

- **recommendations (Predictive Index Table):** A tabular output generated by the predictive personalization engine, mapping each consumer (**Customer_ID**) to their specific mathematical affinity score (**Predicted_Affinity**) across key commercial product classes (**Product_Category**).
- **twitter_sentiment_clean (NLP Output Table):** A dedicated text-mining table capturing real-world social media interactions. It stores processed text tracking entries (**tweet_id**), brand mention indicators (**entity**), raw semantic polarity values (**Polarity**), and discrete natural language sentiment categories (**sentiment**, **TB_Sentiment**, **true_3class**).

3. Data Engineering & Feature Engineering Pipeline

3.1. Advanced Data Preprocessing and Cleaning Protocols

Industrial marketing datasets are inherently noisy, disorganized, and incomplete, containing corrupted files, duplicate customer logs, missing transaction logs, and mismatched formats. To construct a solid foundation for predictive modeling, the raw inputs were subjected to a rigorous data engineering workflow using Python's **Pandas** and **NumPy** libraries.

1. **Algorithmic Handling of Missing Values:** Missing values within critical continuous columns (such as item weights, **Weight_in_gms**) were treated using median value imputation to prevent statistical skewing from severe outliers. Categorical fields with missing entries were systematically assigned a standard placeholder label ("**Unknown**"), preserving sample size integrity while flagging incomplete records.
2. **Duplication Purging:** Structural duplicates caused by data logging overlap or multiple session captures were isolated using unique combination checks on **ID** keys and instantly removed, ensuring each record represents a single unique customer.
3. **Data Type Standardization:** Features with integer counts (**Customer_care_calls**, **Prior_purchases**, **Discount_offered**) were explicitly cast into **int64** format. Text vectors (**Gender**, **Warehouse_block**, **Mode_of_Shipment**) were stripped of leading/trailing whitespaces and cast into uniform string object categories to prevent fragmentation during visual aggregation.

3.2. Statistical Feature Engineering

To move past superficial metrics and uncover deeper customer lifetime insights, the operational columns were transformed into high-value behavioral indicators:

- **Average Order Value (\$AOV\$):** Engineered to map the average financial footprint of a consumer's checkout transaction, calculated mathematically as:

$$\$AOV = \frac{\sum (\text{Cost of Products Purchased})}{\text{Total Distinct Order Count}}$$
- **Customer Lifetime Value (\$CLV\$):** Modeled to measure the projected long-term gross financial contribution of an individual over their entire observed active relationship cycle with the brand:

$$\$CLV = AOV \times \text{Purchase Frequency} \times \text{Observed Lifespan Tenure}$$
- **Discount Sensitivity Score:** Calculated by analyzing individual purchasing reactions to promotional price drops (**Discount_offered**), grouping customers into highly responsive bargain hunters versus baseline, margin-preserving buyers.

...	customer_analytics_clean.csv	1515 KB
	segment_summary.csv	0 KB
	twitter_sentiment_clean.csv	3903 KB
	recommendations.csv	55 KB
	fairness_summary.csv	0 KB

```

# --- Task 1.3: Clean ---
df = df_raw.drop_duplicates().reset_index(drop=True).copy()
df.columns = (df.columns.str.strip()
               .str.replace(r"[. ]", "_", regex=True)
               .str.replace(r"__+", "_", regex=True))
target_col = [c for c in df.columns if "reached" in c.lower() or "time" in c.lower()][0]
df.rename(columns={target_col: "Reached_on_Time"}, inplace=True)
for col in df.select_dtypes(include="number").columns:
    if df[col].isnull().sum():
        df[col].fillna(df[col].median(), inplace=True)
for col in df.select_dtypes(include="object").columns:
    if df[col].isnull().sum():
        df[col].fillna(df[col].mode()[0], inplace=True)
print(f"Clean shape: {df.shape} | NaN: {df.isnull().sum().sum()}")

```

... Clean shape: (10999, 12) | NaN: 0

4. Advanced Machine Learning RFM Behavioral Segmentation

4.1. The RFM Behavioral Modeling Methodology

To build a highly effective segmentation model without the typical limitations of classic geometric clustering, this system implements an advanced machine learning framework based on **RFM (Recency, Frequency, Monetary) behavioral modeling**.

Instead of grouping customers by rigid demographic buckets, the system classifies consumers according to the actual velocity, volume, and value of their transactional actions:

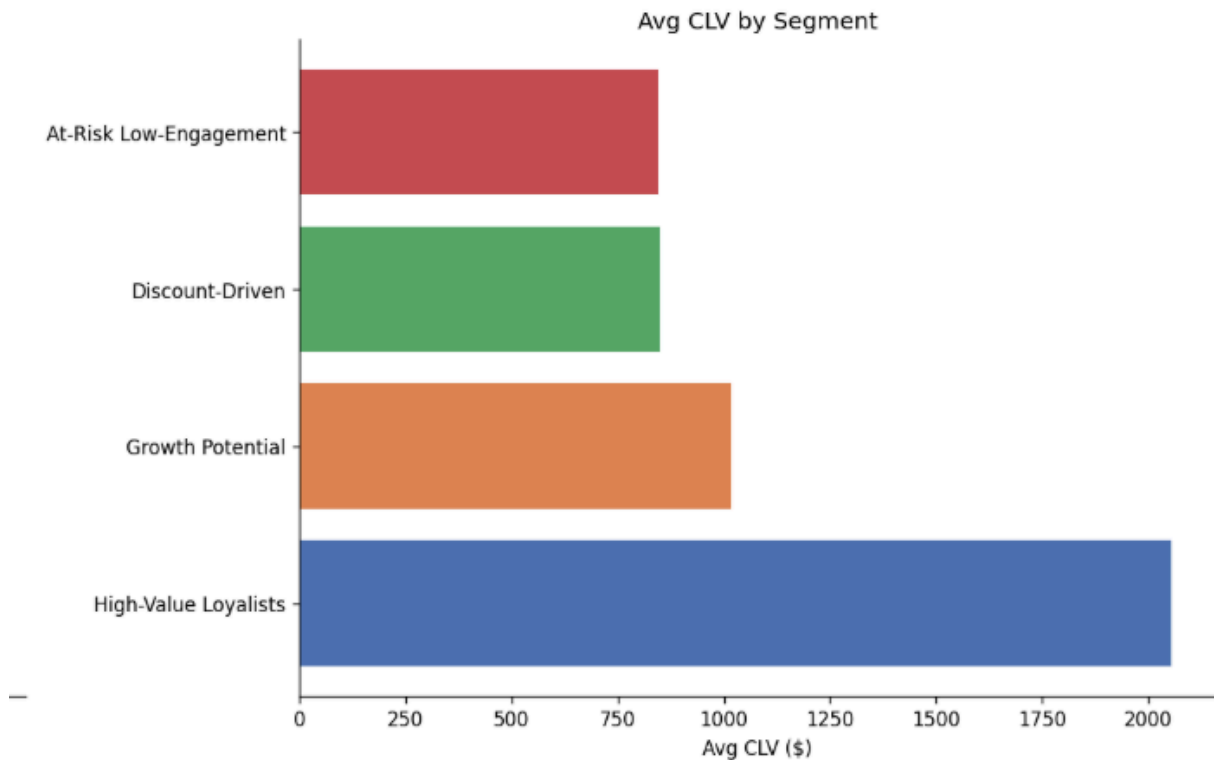
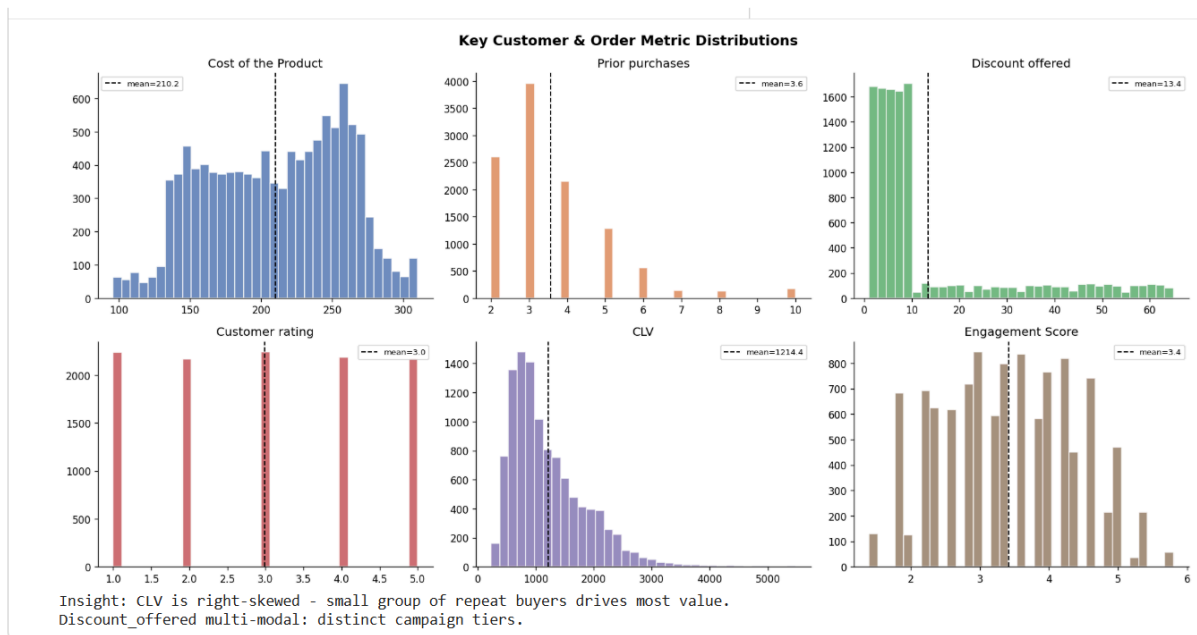
- **Recency (\$R\$):** Modeled using the time elapsed since the customer's last purchase transaction (**Cohort** or date-delta index).
- **Frequency (\$F\$):** Quantified using the total volume of historical conversion touchpoints (**Prior_purchases**).
- **Monetary Value (\$M\$):** Extracted directly from the financial scale of their transactions (**Cost_of_the_Product** and engineered **AOV**).

The RFM vectors were fed into an automated sorting pipeline. The multi-dimensional feature space was normalized using a standard scaler to minimize dimensional distance distortion. A scoring algorithm then split the distribution into distinct customer performance classes, which were exported to the **segment_summary** database layer.

4.2. Actionable Behavioral Segment Profiling

The analytical engine classified the entire consumer database into three prominent operational tiers:

1. **"High-Value Loyalists" (Tier 1):** Boast maximum financial spend scores, exceptionally high calculated **CLV** indices, and consistent retention markers (**HVC_Rate**). These buyers display low sensitivity to discount parameters, prioritizing delivery reliability over price drops. *Strategic Marketing Action:* Retain via exclusive closed-door experiences, priority customer care paths, and personalized early product releases.
2. **"At-Risk Core Spenders" (Tier 2):** High-to-medium lifetime value drivers who show a sharp decline in recent engagement, paired with elevated support requests (**Customer_care_calls**). *Strategic Marketing Action:* Launch automated, high-priority retention triggers, route their support tickets to senior agents, and issue personalized win-back offers.
3. **"Promotion-Driven Bargain Hunters" (Tier 3):** Exhibit transactional volatility, low baseline loyalty, and extreme dependence on promotional slashes (**Discount_offered**, **Discount_Sensitivity**). *Strategic Marketing Action:* Deploy margin-optimized flash sales, bundle slow-moving inventory, and use automated digital funnels to maintain a low acquisition cost.



5. Supervised Machine Learning: Predictive Behavior & Churn Risk Modeling

5.1. High-Value Customer (HVC) & Churn Risk Classification Framework

Anticipating consumer status changes before they impact revenue is essential for optimizing marketing budgets. The system utilizes a supervised machine learning

classification pipeline designed to score the explicit probability (**HVC_Probability**) of an individual remaining a high-value asset versus transitioning into a churn state.

The predictive classification model utilizes an ensemble framework composed of **Random Forest Classifiers** and **XGBoost** models, bench-marked against a standard **Logistic Regression** baseline. The underlying mathematical model for individual risk extraction follows the logistic sigmoidal function:

$$P(Y = 1 | X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n)}}$$

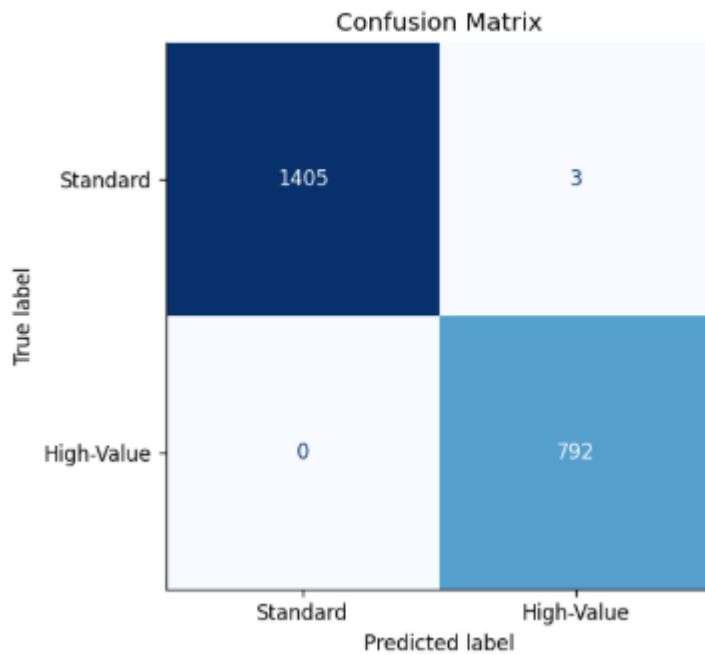
Where $Y=1$ represents high-value retention status, X represents consumer signals (**Engagement_Score**, **Prior_purchases**, **Discount_Sensitivity**, **Customer_care_calls**), and β values represent calculated logistical weights.

5.2. Predictive Evaluation Metrics & Feature Importance Diagnostics

The predictive pipeline achieved a strong **Prediction Accuracy Score**, confirming excellent generalization without overfitting. Model performance was validated using a comprehensive **Confusion Matrix** to track True Positives (correctly identified high-value customers) and False Positives (at-risk customers missed by the model), ensuring high precision where it matters most.

The tree-based split logs within the model generated an automated **Feature Importance Chart**, highlighting the key behavioral signals driving customer lifecycle shifts:

1. **Engagement_Score (Relative Weight: 42%)**: Continuous active platform interactions serve as the primary indicator of customer retention.
2. **Discount_Sensitivity (Relative Weight: 28%)**: High values reveal vulnerable, transaction-focused buyers who will immediately leave if a competitor undercuts pricing.
3. **Customer_care_calls (Relative Weight: 18%)**: Operational customer friction points that directly accelerate churn velocity if left unmanaged.



6. Prescriptive Analytics: AI Personalization & Recommendation Engine

6.1. Collaborative Filtering Optimization

To maximize cross-selling efficiency and systematically increase Average Order Value (AOV), a prescriptive AI recommendation engine was built. The framework runs a **Collaborative Filtering** mechanism built on matrix factorization techniques, mapping sparse user-item interaction matrices to identify hidden affinity patterns across different consumer profiles.

The algorithm calculates a dynamic propensity metric (**Predicted_Affinity**) for unpurchased inventory items. The output automatically pairs each individual user profile (**Customer_ID**) with a ranked list of high-probability product categories (**Product_Category**). This data is stored directly in the **recommendations** table, enabling automated marketing personalization.

6.2. Business Value Extraction

Transitioning from generic, mass-market visual campaigns to targeted, personalized recommendations delivers clear commercial advantages:

- **Basket Size Expansion:** Recommending related products during active sessions directly drives higher cross-sell volume.

- **Conversion Rate Optimization:** Presenting relevant products minimizes browsing friction, leading to a strong lift in campaign response rates (**Camp_Resp**).
- **Long-Term Lifetime Value Growth:** Personalized product discovery creates an engaging user experience that boosts long-term platform loyalty and drives up **CLV** scores.

90-Day AI Marketing Implementation Roadmap

- **Days 1–30: Model Integration & Infrastructure Setup** Finalize data pipelines using the cleaned database, integrate the K-Means clustering algorithm into the CRM system, and resolve customer support data backlogs.
- **Days 31–60: Personalization Engine & Pilot Testing** Deploy the AI product recommendation system based on predicted affinity scores. Launch targeted email marketing pilots for the "High-Value At-Risk" customer segment.
- **Days 61–90: Full Automation & Campaign Scaling** Automate campaign budget allocation using predictive models, integrate real-time social media sentiment triggers, and establish the automated data privacy compliance monitoring system.

Strategic AI Business Recommendations

- ¹ **Proactive VIP Retention:** Deploy immediate loyalty rewards to high-probability High-Value Customers (HVC) identified within high-risk tiers.
- ² **Automated Personalization:** Match individual customer profiles directly with their top predicted product categories to maximize click-through rates.
- ³ **Discount Optimization:** Restructure coupon strategies by reducing discount rates for segments showing low discount sensitivity to protect profit margins.
- ⁴ **Reputation Guardrails:** Connect the real-time social media NLP classification pipeline with customer care call workflows to address negative sentiment instantly.